

SatQuery AI

Evidence-backed remote sensing answers through natural language

Official template fields

PROBLEM STATEMENT ID	SIH26167
PROBLEM STATEMENT TITLE	SatQuery AI - An Interactive Vision-Language Assistant for Multimodal Remote Sensing Image Analysis through Text Queries
THEME	Space Technology
PS CATEGORY	Software
TEAM ID	[REGISTERED TEAM ID]
TEAM NAME	[REGISTERED TEAM NAME]

Core promise

Ask a question about satellite imagery. The system validates the data, selects the correct specialist model, and returns a spatial answer with evidence.

Replace the two bracketed placeholders before submission.

Proposed Solution

One question. The right specialist model. A spatial answer that can be verified.

What the system does

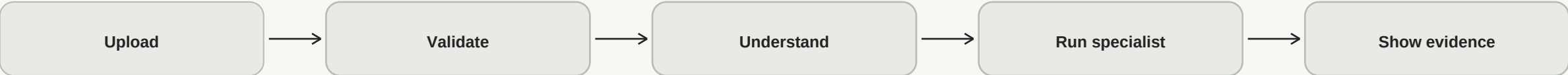
- Accepts a single scene, a bi-temporal pair, or a co-registered optical-SAR pair with a natural-language question.
- Validates file, CRS, bands, metadata, alignment, and image quality before inference.
- Returns a grounded answer with map or mask, area, coordinates, confidence, model version, and trace.

How it solves the problem

- Combines validation, model selection, analysis, and evidence delivery in one guided workflow.
- Uses task-specific models instead of one generic vision-language model for every request.
- Converts technical model outputs into understandable spatial evidence.

Why it is different

- Sensor-aware routing across optical, SAR, and temporal imagery.
- Team-developed SAR-FuseSeg for spatial optical-SAR fusion.
- Evidence before language, calibrated confidence, visible warnings, and safe abstention.
- Public benchmarks isolated for evaluation only.



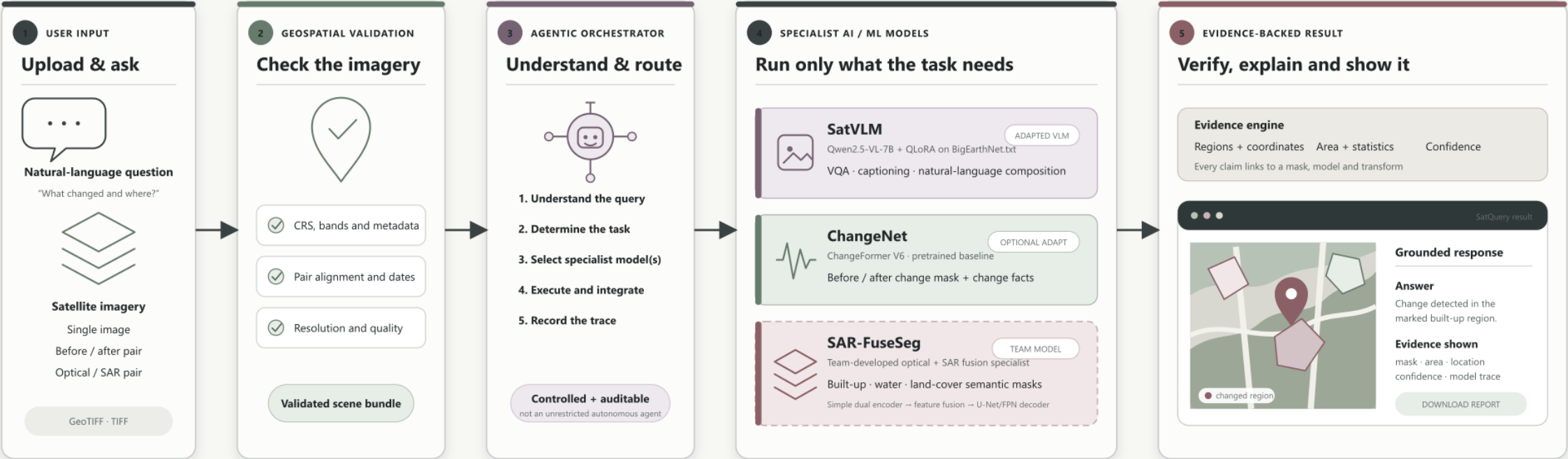
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JUDGE PRESENTATION VIEW

From Satellite Imagery to an Evidence-Backed Answer

A five-step agentic workflow that chooses the right specialist, verifies its evidence and explains the result on a map.

Detailed engineering architecture remains in the appendix



ARCHITECTURE PRINCIPLE

Specialist models create masks and measurements. SatVLM converts only verified facts into natural language.

WHY JUDGES CAN TRUST IT

Visible model choice · calibrated confidence · evidence overlays · execution trace

Training: BigEarthNet.txt for SatVLM · BigEarthNet v2 + DFC2020/SEN12MS for SAR-FuseSeg

Evaluation-only: VRSBench · RSVQA · CDVQA · hidden ISRO/SAC

● adapted VLM ● change specialist ● team-developed model

Pitch line: "The system chooses the right specialist and returns evidence, not unsupported AI text."

Feasibility and Viability

Why this is feasible

- Public datasets and pretrained baselines cover every core task.
- Only SatVLM QLoRA and SAR-FuseSeg require planned training.
- A modular design allows independent testing, replacement, and deployment.
- Six students, eight weeks, and about 50 to 90 L40S GPU-hours form a realistic prototype plan.

Challenge or risk	Mitigation strategy
Sentinel to Cartosat or RISAT domain shift	Sensor-specific preprocessing, augmentation, multi-dataset stress tests, and held-out calibration
Benchmark leakage or inflated scores	Evaluation-only benchmark vault with frozen split IDs, hashes, and access logs
Misaligned image pairs	CRS, bounds, resolution, overlap, date, and co-registration checks before inference
Unsupported answers	Compose language only from verified evidence and abstain below the task threshold
Different confidence scales	Calibrate each specialist separately and never average raw model scores
Limited compute and time	Use 4-bit QLoRA, mixed precision, tiling, pretrained baselines, and smoke tests

EIGHT-WEEK DELIVERY PATH Data and baselines -> Model adaptation -> Integration -> Calibration, stress testing, offline demo, and freeze

Impact and Benefits

Target audience

ISRO, SAC, and NRSC analysts; disaster management agencies; agriculture, forestry, water, infrastructure, and urban-planning departments; researchers and policy teams.

Potential impact

- Natural-language access to complex satellite analysis.
- Trust through maps, masks, coordinates, confidence, warnings, and model traces.
- Faster first-level interpretation across optical, SAR, and temporal imagery.
- Secure or offline deployment for sensitive imagery.

Social

- Faster evidence for disaster response and public-service planning.
- Easier access for users with limited GIS or ML expertise.
- Traceable outputs support responsible human review.

Economic

- Can reduce repetitive manual inspection and separate-tool dependence.
- Reuses pretrained models and public data to lower prototype cost.
- Prioritizes regions before expert review or field verification.

Environmental

- Supports monitoring of land cover, water, forests, and urban expansion.
- Combines optical context with SAR evidence under cloud or low light.
- Produces spatial outputs that can be compared over time.

Research and References

Primary sources for the problem, model choices, datasets, and evaluation plan.

1. Smart India Hackathon 2026, Problem Statement SIH26167

<https://sih.gov.in/sih2026PS#ViewProblemStatement26167>

2. Qwen2.5-VL-7B-Instruct model card

<https://huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct>

3. BigEarthNet.txt project and paper

<https://txt.bigearth.net/>

<https://arxiv.org/abs/2603.29630>

4. ChangeFormer paper and repository

<https://arxiv.org/abs/2201.01293>

<https://github.com/wgcban/ChangeFormer>

5. BigEarthNet v2

<https://bigearth.net/>

6. IEEE GRSS Data Fusion Contest 2020

<https://www.grss-ieee.org/community/technical-committees/2020-ieee-grss-data-fusion-contest/>

7. VRSBench

<https://arxiv.org/abs/2406.12384>

<https://vrsbench.github.io/>

8. RSVQA

<https://arxiv.org/abs/2003.07333>

9. CDVQA

<https://arxiv.org/abs/2112.06343>

Team contribution statement

SAR-FuseSeg is our proposed team-developed optical-SAR semantic segmentation model. The cited datasets and baseline research support its design, training, and evaluation plan.